

Building Your AI-Ready Curriculum

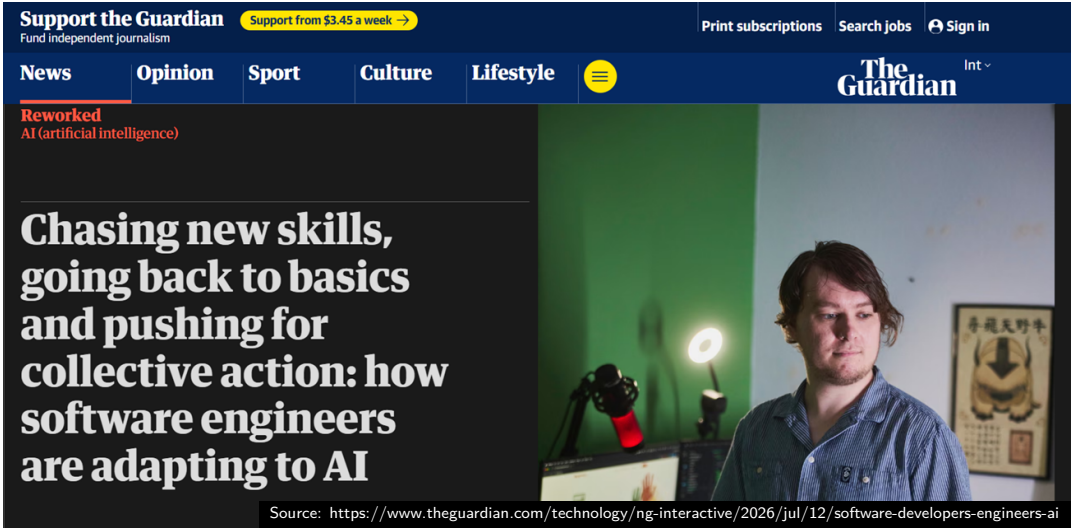
A Practical Toolkit for Programme Design and Review

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Impact of AI on Professional Practice (1)



Impact of AI on Professional Practice (2)

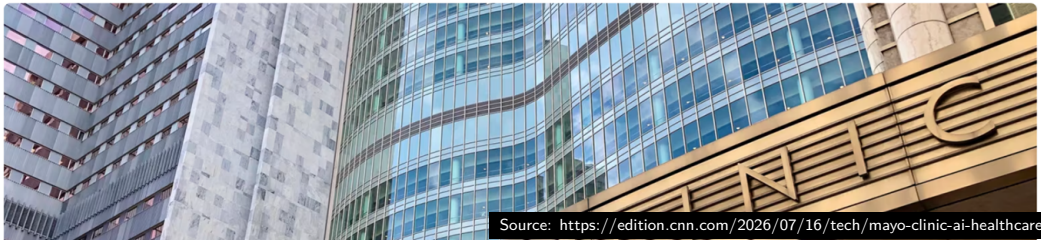
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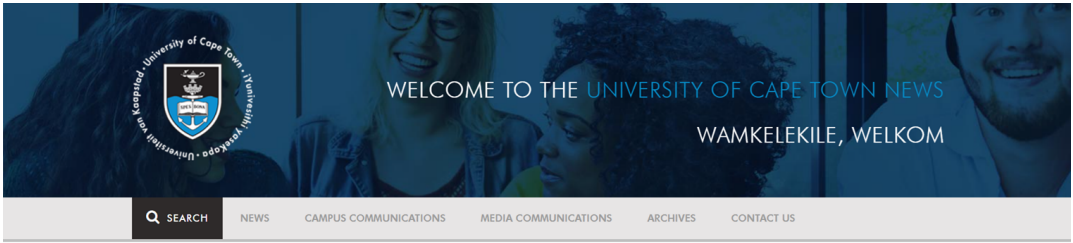
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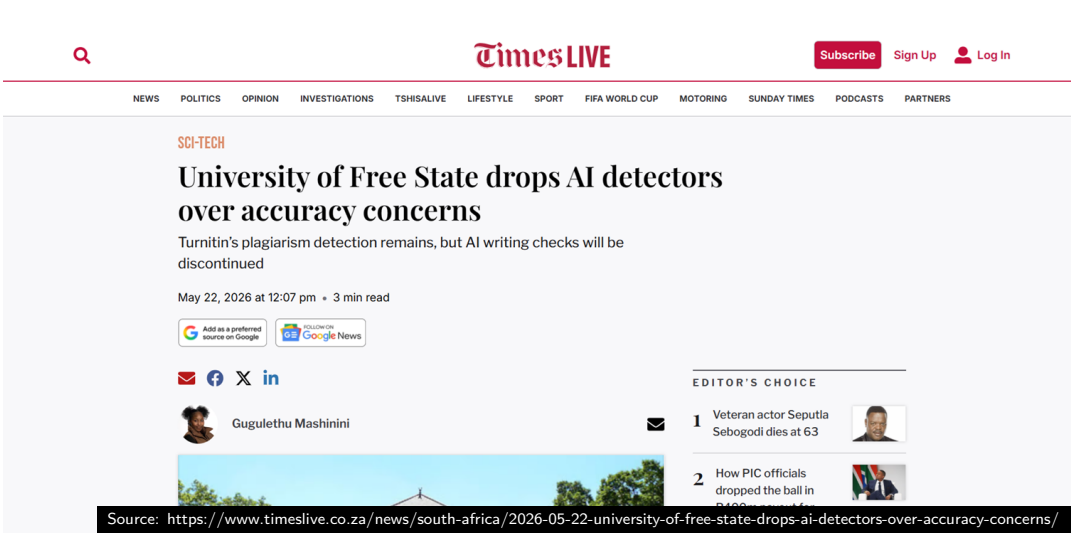
Source: https://edition.cnn.com/2026/07/16/tech/mayo-clinic-ai-healthcare

Impact of AI on Professional Practice (3)



Source: https://www.news.uct.ac.za/article/-2025-07-24-uct-scraps-flawed-ai-detectors

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AI is Changing Every Profession

Artificial Intelligence is no longer confined to Computer Science.

It is reshaping how professionals work across every discipline.

Examples

- Education
- Medicine
- Law
- Engineering
- Agriculture
- Business

AI is changing...

- Decision-making
- Data analysis
- Content creation
- Communication
- Research
- Professional practice

Why Curriculum Review Must Change

The pace of technological change now exceeds the pace of curriculum review.

The Traditional Model

- Curriculum reviewed every 4–5 years
- Focus on updating content
- Compliance-driven processes
- Slow response to industry change
- Graduates risk leaving with outdated skills

The AI Era

- AI capabilities evolve every few months
- Professional practice changes continuously
- New tools emerge rapidly
- Employer expectations shift quickly
- Lifelong learning becomes essential

Traditional vs AI-Ready Curriculum Review

Traditional Review	AI-Ready Review
Every 4–5 years	Continuous environmental scanning
Content-centred	Graduate capability-centred
Compliance-focused	Innovation and improvement focused
Knowledge acquisition	Knowledge, skills and judgement
Static programme design	Adaptive programme design
Limited employer engagement	Ongoing collaboration with industry
Assessment of recall	Assessment of authentic professional performance

What is an AI-Ready Curriculum?

Definition

An AI-ready curriculum equips graduates to work effectively, critically, ethically, and adaptively in a world where Artificial Intelligence is increasingly integrated into professional practice.

An AI-ready curriculum is NOT...

- A curriculum that teaches only AI.
- A curriculum that replaces human expertise.
- A curriculum where every course uses AI.
- A curriculum that depends entirely on AI tools.

An AI-ready curriculum IS...

- Preparing graduates for AI-enabled workplaces.
- Developing critical and ethical AI users.
- Integrating AI where it genuinely adds value.
- Building adaptable lifelong learners.

The Five Pillars of an AI-Ready Curriculum

AI-Aware	Understands AI concepts, capabilities and limitations.
AI-Capable	Uses AI tools productively within the discipline.
AI-Critical	Evaluates, verifies and questions AI-generated outputs.
AI-Ethical	Applies AI responsibly with consideration for privacy, bias, transparency and academic integrity.
AI-Adaptable	Continues learning as AI technologies evolve throughout a professional career.

Think of these as graduate capabilities, not individual courses.

Every programme should determine how these capabilities are developed within its own disciplinary context.

Common Misconceptions About AI-Ready Curricula

Misconception	Reality
"We need to add an AI course."	AI readiness is integrated across the curriculum, not confined to a single module.
"We should ban AI completely."	Students need to learn how to use AI responsibly and critically, just as they learn to use other professional tools.
"AI will replace lecturers."	AI complements teaching by supporting learning, feedback and assessment, while lecturers remain central to designing meaningful learning experiences.

A Practical Toolkit for Building AI-Ready Curricula

A structured approach to programme design and review

1. Scan the Environment
2. Review Graduate Attributes
3. Audit the Curriculum
4. Revise Learning Outcomes
5. Redesign Assessment
6. Build Staff Capability
7. Continuous Quality Improvement

Step 1: Scan the Environment

Before changing the curriculum, understand how the profession is changing.

Why begin here?

Curriculum review should be driven by evidence rather than assumptions. An environmental scan helps institutions identify the trends, technologies and expectations that are reshaping professional practice.

Questions to Ask

- What has changed in our profession?
- Which AI tools are becoming common?
- What tasks are becoming automated?
- What new competencies are emerging?

Key Principle

- Don't begin with modules.
- Don't begin with assessments.
- Begin with the changing world of work.

Step 2: Review Graduate Attributes

What capabilities should every graduate possess in an AI-enabled world?

Graduate attributes are the bridge between the curriculum and professional practice.

Rather than asking *"What should we teach?"*, ask:

"What kind of professional do we want to graduate?"

Traditional Focus

- Subject knowledge
- Technical competence
- Communication
- Teamwork & Problem solving

Expanded for the AI Era

- AI literacy & Critical evaluation
- Ethical judgement
- Human-AI collaboration
- Adaptability and lifelong learning

Step 2 (Continued): From Graduate Attributes to Curriculum Design

Graduate Attribute	Possible Curriculum Implications
Critical AI use	Students evaluate AI-generated outputs rather than simply accepting them.
Ethical judgement	Include discussions of bias, privacy, transparency and accountability.
Adaptability	Provide opportunities to learn unfamiliar tools independently.
Professional communication	Require students to explain and justify decisions made with AI support.
Collaboration	Design activities where students work effectively with both peers and AI tools.

Step 3: Audit the Curriculum

Understand where your programme already supports AI readiness, and where the gaps are.

Curriculum auditing is about evidence, not assumptions.

Rather than immediately introducing new courses or content, examine how the existing curriculum contributes to the graduate attributes identified in the previous step.

Review Existing Modules

- Learning outcomes
- Topics covered & Assessments
- Teaching approaches
- Graduate capabilities developed

Look for...

- Missing capabilities
- Duplication & Outdated content
- Opportunities for AI integration
- Better alignment across modules

Step 4: Revise Learning Outcomes

Learning outcomes define what graduates should know, do and value.

Why revise learning outcomes?

If graduate capabilities have changed, then programme and course learning outcomes should reflect those changes. Well-designed learning outcomes become the foundation for teaching, learning activities and assessment.

Traditional Outcomes

- Focus on knowledge acquisition
- Emphasise technical competence
- Often discipline-specific
- Limited reference to AI

AI-Ready Outcomes

- Critical evaluation of AI outputs
- Ethical and responsible AI use
- Human-AI collaboration
- Adaptability and lifelong learning

Step 5: Redesign Assessment

If AI can complete an assessment with little human thinking, what are we really assessing?

The Challenge

Generative AI has made it possible for students to produce essays, reports and code with unprecedented ease. Rather than asking how to prevent AI use entirely, we should ask how to assess the capabilities that matter most.

Assess Less Of...

- Recall of facts
- Generic essays
- Repetitive reports
- Tasks with a single correct answer

Assess More Of...

- Critical reasoning & Decision-making
- Reflection
- Professional judgement
- Application to authentic contexts

Step 5 (Continued): Designing Authentic AI-Ready Assessments

Authentic assessment mirrors the kinds of tasks graduates will encounter in professional practice.

Traditional Assessment	AI-Ready Alternative
Essay	Case study with justification of decisions
Programming assignment	Develop a solution and explain design choices, including any AI assistance used
Research report	Critique AI-generated summaries and synthesise evidence independently
Presentation	Present findings and defend conclusions during questioning
Written examination	Scenario-based analysis requiring professional judgement

Step 5 (Continued): Academic Integrity in the Age of AI

Integrity is strengthened through transparency, not uncertainty.

Encourage Students To...

- Declare AI use where appropriate
- Verify AI-generated content
- Cite sources accurately
- Reflect on how AI informed their work
- Accept responsibility for the final submission

Institutions Should...

- Develop clear AI policies
- Provide consistent guidance
- Design authentic assessments
- Support staff development
- Promote ethical AI use

Step 6: Build Staff Capability

An AI-ready curriculum requires AI-ready educators.

Why staff capability matters

Curriculum transformation depends not only on programme design, but also on the confidence and competence of academic staff. Lecturers need opportunities to explore AI, share effective practices and develop new pedagogical approaches.

Knowledge

- AI concepts and terminology
- Opportunities and limitations
- Discipline-specific applications
- Ethical considerations

Skills

- AI-enhanced teaching
- Assessment redesign
- Prompt engineering
- Critical evaluation of AI outputs

Step 7: Continuous Quality Improvement

AI-ready curricula evolve continuously, not just during formal review cycles.

From Periodic Review to Continuous Enhancement

Artificial Intelligence is changing rapidly. To remain relevant, programmes should adopt an evidence-based approach to continuous monitoring and improvement rather than waiting for the next scheduled curriculum review.

Traditional Approach

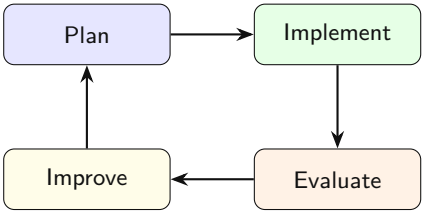
- Review every 4–5 years
- Major curriculum overhaul
- Reactive updates
- Compliance-driven

AI-Ready Approach

- Annual programme reflection
- Incremental improvements
- Evidence-informed decisions
- Continuous stakeholder engagement

Step 7 (Continued): A Continuous Improvement Cycle

Quality assurance should be an ongoing cycle rather than a compliance exercise.



Sources of Evidence

- Student feedback
- Graduate and alumni feedback
- Employer input
- External examiner reports
- Professional body guidance
- Emerging AI developments

Five Takeaways for Building AI-Ready Curricula

1. **AI readiness is about graduate capabilities, not simply teaching AI.**

2. **Begin with industry needs and graduate attributes before revising courses.**

3. **Redesign learning outcomes and assessments to emphasise critical thinking, judgement and ethical AI use.**

4. **Invest in academic staff development and institutional AI policies.**

5. **Treat curriculum review as a continuous quality improvement process rather than a periodic compliance exercise.**

"The question is no longer whether AI belongs in our curricula, but how we prepare graduates to thrive in an AI-enabled world."

Reflection

Building AI-Ready Programmes

Every institution is at a different stage of its AI journey. The goal is not to have all the answers today, but to begin asking the right questions during programme design and review.

Reflection Questions

1. What is the greatest opportunity AI presents for curriculum design in your institution?
2. What is the greatest challenge you anticipate when implementing AI-ready curricula?
3. Which element of the AI Curriculum Review Canvas could your programme begin applying immediately?

Thank You

Questions